

[INTERNAL] Stacks blockchain integration with Fireblocks using RAW Signing

Before you begin

To use either the SDK or the desktop app solution, you must have Raw Signing enabled in your Fireblocks workspace. Please contact your Customer Success Manager for more information.

Overview

Fireblocks does not currently support Stacks blockchain natively. This guide outlines a dedicated solution for managing Stacks operations with Fireblocks, utilizing Fireblocks' RAW Signing capability alongside a purpose-built SDK and desktop application.

This solution abstracts the complexity of Stacks operations, allowing developers and users to effortlessly initiate funds transfers, monitor balances, manage staking operations, and track transaction history with minimal implementation overhead.

Warning

Fireblocks RAW signing has limited TAP capabilities hence the user needs to take all required measurements to reduce any security risk. Contact the Fireblocks team to get information about RAW signing security best practices

Our solutions

The Fireblocks solution offers two options for interacting with the Stacks blockchain, Both options share the same prerequisites and infrastructure, and the Desktop UI interacts with the same SDK backend, supporting identical features.

Prerequisites

- Fireblocks RAW signing feature enabled
- Fireblocks API user with appropriate permissions
- Transaction Authorization Policy (TAP) rule in place for RAW signing
- **Optional:** Hiro platform API key, optional yet recommended as it significantly improves API rate limits and overall user experience, can be obtained for free by registering to [Hiro Platform](#)

Option 1: SDK with REST API

The Stacks-Fireblocks SDK can be used in two ways: directly as a TypeScript library by initializing it and calling its methods programmatically, or as a REST API by interacting with the provided HTTP endpoints - offering flexibility for both integrated and service-based use cases.

The Stacks-Fireblocks SDK enables developers to:

- Fetch the Stacks account addresses for a Fireblocks vault account.
- Retrieve Stacks account balances for both native STX and custom tokens.
- Get detailed information on all tokens owned by the account.
- Create Stacks transactions - the SDK builds the transaction, sends it for signing in your Fireblocks account, and submits it to the blockchain automatically.
- Retrieve detailed transaction history for all transactions sent and received by the account.

- Replace stuck or pending transactions by resubmitting them with a higher fee (replace-by-fee / RBF) — supporting both explorer-visible transactions reconstructed automatically by the SDK, and unindexed transactions using a direct nonce override.
- Manage the complete Stacks staking (stacking) lifecycle:
 - Submit solo stacking request to POX (Proof Of Transaction) contract
 - Delegate STX to the supported stacking pools
 - Submit requests to increase locked funds amount or extend the locking period
 - View stacking status for vault account

Option 2: Desktop UI Application

Recommended for non-technical users, this Electron-based local application provides an intuitive interface to:

- Set up Fireblocks API credentials securely with guided CSR generation
- View vault accounts with Stacks addresses and balances
- Create and sign Stacks transactions (vault-to-address or vault-to-vault)
- Monitor transaction history
- Manage complete stacking lifecycle through dedicated modals
- View stacking information, delegation status, and rewards
- Browse all Stacks-native DeFi applications and stacking pools — DEXs, lending protocols, and liquid staking — and interact with them directly using your Fireblocks vault as the connected wallet
- Authenticate securely with biometric (Touch ID on Mac) or password-based authentication

Installation & Setup

1. Enable Raw Signing

To request the activation of this feature, please reach out to your Customer Success Manager. Please note that the Raw Signing capability is classified as a premium feature and may incur an additional platform fee if enabled.

Additionally, it is essential to recognize that the Raw Signing feature can introduce potential security vulnerabilities. You must strictly follow all Fireblocks security guidelines and best practices when utilizing RAW signing to minimize associated risks.

Learn more about Fireblocks Raw Signing [here](#). For detailed information or assistance, please contact the Fireblocks team at any time.

2. Configure the Policy

To submit Stacks transactions on the blockchain, transactions must be structured and signed according to the requirements set by the network. The SDK or Desktop application takes care of generating these transactions and submitting them to your Fireblocks workspace for signing.

For all Stacks transactions and operations, the signature request will use the Raw Signing operation.

To enable this signing operation, your workspace must have a dedicated Policy rule in place for this signing method to ensure the signing process can proceed successfully.

Raw Message Signing TAP rule

- All Raw Signing transactions are blocked by default. You must create Policy rules to explicitly allow the Raw transaction type.
- You should specify who can initiate and who can approve Raw transactions. Best practice is to require a two-tier setup (initiator and a human approver) or use an API user plus a human validator. For better security, avoid initiator rules with Any as the initiator when possible.
- Limit rules to only the source vault accounts, and ideally, the specific assets and derivation paths needed for your workflow.

- Raw Signing rules do not support amount caps or destination controls. This is a major risk compared to normal transfer/contract call rules.
- Raw Signing should be tightly scoped and temporary where possible. Remove or update RAW rules immediately after use if the need is one-off.
- Always set up multi-approver or designated signer flows for additional oversight.
- Avoid using the **Any vault** or **Any user** values unless necessary. More granular = more secure.
- Educate all users on the risks. Raw Signing can bypass most usual controls and is equivalent to signing a “blank check.”
- Consider using an API Co-signer with a Callback Handler for programmatic and additional business logic validation.

3. SDK setup

To get started with the SDK, clone our public repository.

The repository's README.md file contains all setup and deployment instructions, as well as TypeDoc documentation. Follow those steps to complete your local or Docker-based environment.

4. Using the desktop app

The Fireblocks x Stacks Desktop Application provides a full-featured, user-friendly interface for managing Stacks operations.

Designed for users who prefer not to work with code, the application offers secure local access to your Fireblocks vaults and transaction signing workflows via Raw Signing. It includes biometric authentication, STX and Custom Tokens transferring, STX stacking (Solo stacking and delegation to stacking pools via wallet-connect), transaction history monitoring and real-time updates of balances and staking status.

Key features

- **Secure Authentication:** Uses your system's biometric authentication (Touch ID on Mac) or password-based login (Windows/Linux) to protect access. Credentials are stored securely in the OS keychain.

- **API Credential Setup:** Guided setup process with options to generate a new Certificate Signing Request (CSR) or import existing Fireblocks API credentials.
- **Vault Account Management:** View, create, and manage Stacks-enabled vault accounts with real-time balance updates.
- **Funds Management:** Check balances and transfer Native STX and Custom Tokens securely.
- **Replace By Fee:** Re-submit pending/stuck transactions with higher fee to allow them to be mined faster.
- **Transaction History:** View detailed transaction records per vault account.
- **Stacks Staking Management:**
 - Submit solo stacking request to POX (Proof Of Transaction) contract
 - Delegate funds to Stake Pools
 - Submit requests to increase locked funds amount or extend the locking period
 - Real-time staking status and rewards tracking

Getting started

Download the app

Download the app from our GitHub repo [here](#).

First launch

Double-click the Fireblocks x Stacks application icon.

When prompted, authenticate using Touch ID (on Mac) or your system's configured authentication method (password on Windows/Linux).

After the welcome screen introduces the application, select Get Started to begin setting up your credentials.

If this is your first time opening the app after downloading it directly from the developer, you may need to right-click the app icon and select Open to bypass security warnings on macOS.

Setting up your Fireblocks API keys

The application requires Fireblocks API credentials to function (connects to your Fireblocks account).

On the API Key Setup screen, you can:

1. **Generate a new Certificate Signing Request (CSR):** The app guides you through generating an RSA key pair and CSR locally. Download the CSR file, upload it to your Fireblocks Console (Settings → API Users), and retrieve your API key. Paste the API key back into the app to complete setup.
2. **Use existing credentials:** If you already have a Fireblocks API user and secret key file, you can import them directly. Browse to select your secret key file and enter your API key.

Once your credentials are validated, the app will securely store them in the system's keychain. You will then be redirected to the Secondary Credential Setup.

Setting up your Hiro Platform API key

1. The Hiro API key is optional and this step can be skipped, however it is highly recommended as it improves rate limits and overall user experience significantly.
2. Enter your Hiro API key in the Secondary Credential Setup screen.
3. The app will validate the key by performing a health check.

Once validated, the key is securely stored in your system keychain.

You will be redirected to the main dashboard after successful setup.

Main Dashboard

The main dashboard displays all your Fireblocks Vault accounts with Stacks capability, showing Stacks balances (STX, USDCx & sBTC).

Key elements

- **Header:** Contains the app title, refresh button, notification indicator, and Settings button.
- **Search bar:** Filter Vault accounts by name or ID.

- **Refresh button:** Refresh the list of Vault accounts and update all Stacks balances.
- **Create button:** Add new Vault accounts with automatic Stacks wallet creation.
- **Vault Account list:** Shows your Vault accounts with their balances.

Viewing Vault account details & Transaction history

The Vault Accounts table shows each Vault account's name, ID, and Stacks balances.

Select any Vault account to view its details page, which includes:

- Account information
- Vault account name and ID
- Stacks address and corresponding BTC rewards address
- Total STX balance
- All owned custom tokens and balances
- Stacking information
 - Current delegation status
 - Total funds locked
 - Lock period duration

From a Vault account's details page, you can:

- **Create Transfer:** Send STX and custom tokens to another address or vault account
- **Stacking Operations:**
 - **Solo Stacking:** Submit lock funds request
 - **Increase amount:** Submit request to increase locked funds amount
 - **Extend period:** Submit request to extend funds lock period
- **Transaction History:** View detailed transaction history for both transfers and contract calls with 'View on Explorer' option

- **Replace Pending Transaction:** Resubmit a stuck pending transaction with a higher fee to accelerate confirmation, pending transactions will also appear in Transaction History page with replacing option (opens RBF modal pre-filled with transaction details)

Creating Stacks transactions

To send funds:

1. Go to the appropriate Vault account's details page.
2. Select **Create Transfer**.
3. On the transaction modal, configure:
 - a. **Token:** Choose the Stacks token to transfer
 - b. **Destination:** Enter a recipient Stacks address OR select a vault account from the drop-down
 - c. **Amount:** Enter amount to send, or toggle Use Max Amount to send all available tokens
 - d. **Fee handling:** Choose gross transaction (deduct fee from amount) or leave blank for net transaction (add fee to amount)
4. Review the transaction summary, then select Create Transaction to initiate the transfer.
5. Authenticate with Touch ID (Mac) or password (Windows/Linux) when prompted.
6. Sign the transaction using the Fireblocks mobile app or automatic cosigner per your workspace policy.
7. After confirmation, the transaction will appear with **processing** status until it's confirmed on the Stacks blockchain.

Stacking Operations

To lock (stack) STX:

8. Go to the appropriate Vault account's details page.
9. Select **Solo Stack**.
10. On the stacking modal, configure:
 - a. **Action:** Stack STX, increase amount, or extend period
 - b. **Parameters:** Enter the parameters relevant to your action
 - c. **Signer key and signature:** Use [Stacks Stacking Signature Tool](#) to get a signer key and generate the signature over the relevant parameters
11. Review the operation summary, then select submit to initiate.
12. Sign the transaction using the Fireblocks mobile app or automatic cosigner per your workspace policy.
13. After confirmation, the transaction will appear with **processing** status until it's confirmed on the Stacks blockchain.

DApp Browser

To interact with DeFi applications and stacking pools:

1. Go to **DeFi & Staking** in the sidebar.
2. Browse the available applications across two categories:
 - **DeFi Apps:** DEXs, lending protocols, liquid staking, bridges, and other Stacks-native applications.
 - **Stacking Pools:** Non-custodial delegation pools for earning BTC or STX rewards.
3. Select the application or pool you want to use, then select **Open DApp**.
4. The DApp opens inside the application in an embedded browser.
5. Connect your wallet by selecting **Connect Wallet** within the DApp — your Fireblocks vault accounts will be available as the wallet options.

6. Interact with the DApp as normal. Any transaction it requests (contract call, token transfer, etc.) will be routed through Fireblocks and prompt the standard signing flow.
7. Confirm the operation with the confirmation modal.
8. Sign the transaction using the Fireblocks mobile app or automatic cosigner per your workspace policy.
9. After confirmation, the transaction will appear with processing status until it's confirmed on the Stacks blockchain.

Settings & Security

To access the app's settings, select the gear icon in the bottom-left corner of the app.

Security settings

- **Authentication Lock Time:** Set how long the app stays authenticated before requiring reauthentication. Enter the number of minutes (1-60), then select **Save** to apply the new timeout value. The next time your session expires, you'll have to reauthenticate with Touch ID.

Credentials management

- **Change Fireblocks API Credentials:** Remove the current credentials and set up new ones. This will require re-authentication.
- **Change Hiro API Key:** Update your Hiro API key for Stacks operations.
- **Clear All Data:** Remove all stored credentials and reset the app to initial setup state.

Troubleshooting

Some common issues with the Fireblocks x Stacks desktop app can include the following.

Authentication problems

- **Touch ID not working (Mac):** Ensure Touch ID is properly set up in System Preferences and that the app has permission to use it.
- **Authentication timeout too short:** Increase the timeout value in Settings.
- **Balance not updating:** Select the Refresh button on the Vault accounts list or the Transaction History.
- **Pending transactions:** Some transactions may take time to be confirmed on the blockchain.
- **Transaction failed:** Common reasons include:
 - Insufficient STX balance for transaction fees
 - Insufficient token balance
 - Network congestion
- **Staking issues:**
 - Failing to submit lock funds request: the SDK is designed to reject requests in certain 'Danger Zone' time windows where the network will not be able to process the request and lock funds on time, in this case the user will have to wait until next stacking cycle (could take up to few days).
 - Signature error: please double check the provided signature is generated over parameters that match the parameters provided in the stacking modal.
- **Connection issues:** Check your internet connection and restart the app.

API connection errors

- **Invalid API key:** Verify your Fireblocks API key is correct and has the appropriate permissions for Raw Signing.
- **Invalid Hiro API key:** Check that your Hiro API key is valid. You can update it in Settings.

Getting support

If you continue to experience one or more of these issues after troubleshooting, contact your Fireblocks Customer Success Manager and ask for assistance from the Professional Services team.